

1 Core Architecture: The “Sequence Logic Engine”

At its foundation, Archangel is a **grid-sequence EA** — not a random scalper.

It opens *a series of trades (“sequence”)* spaced apart by **Pip Step**, each sized by **Lot Exponent**, then closes them *collectively* once the average position hits **Take Profit** or **Lock Profit**.

Key mechanics:

Setting	Function	Principle
Allow New Sequence	Controls whether new grids start after one closes.	Like a “main power switch.” FALSE = finishes open series, then stops.
Magic Number	Unique trade ID used to track its own positions.	Prevents conflicts between multiple EAs.
Use Random Entry Delay	Randomizes entry times slightly.	Avoids broker detection / duplicated trades across accounts.
ATR Usage Info / ATR Period	If TRUE, pip distances become <i>adaptive</i> (volatility-scaled).	E.g. ATR = 20 pips → Pip Step = $2.5 \times \text{ATR} = 50$ pips.
Pip Step	Distance between each trade.	Defines the grid density. Larger = lower DD / slower closure.
Pip Step Exponent	Multiplies step size progressively.	Makes later trades spread out farther → less clustering risk.
Max Pip Step	Hard cap on spacing.	Prevents grid from expanding infinitely.

 Think of this block as the “**grid geometry**” — how wide and how dense your trade web is.

2 Sequence Timing & Trade Scheduling

These control **when** trades occur — very useful for fine-tuning risk and “appearance.”

Setting	Function	Analogy
Delay Trade Sequence	Skips the first few trades in a grid.	Like saying “start trading only after the first 3 grid levels.”

Live Delay	Delays multiple entries and executes them all together.	Used to bunch orders for better averaging.
Lot Multiplier for 1st Trade after LD	Boosts the “catch-up” trade size.	Gives more weight to the delayed group’s first real trade.
Combine Live Delay Trades	Merges multiple delayed positions into one.	Reduces trade count; cleaner logs.
Trade Direction	Long, Short, or Both.	Defines directional bias of the algorithm.
Max Orders per Direction	Caps open positions.	The <i>hard stop</i> for exposure in one trend.
Reverse Sequence Direction	Flips logic (Buy→Sell).	Useful for anti-correlation experiments.

🧠 This block defines the “**temporal rhythm**” of your system — how fast and how many positions deploy.

💰 3 Money Management & Exit Logic

Where 80% of your *risk/reward* control happens.

Setting	Concept	Explanation
Take Profit (pips / ATR multiple)	Defines where the full sequence closes.	Based on <i>weighted average price</i> of open trades.
Stop Loss (pips / ATR multiple)	Hard stop for individual trades.	Mostly inactive unless manually enabled.
Lock Profit Min Trades	Minimum trades before trailing activates.	Avoids early trailing before full sequence forms.
Lock Profit (pips)	Distance beyond breakeven where trailing starts.	Once sequence = +X pips → activate lock.
Lock Profit – When to Check	Every tick / Bar Close / M1.	Frequency of trailing evaluation.
Trailing Stoploss (pips)	Distance between price and trailing level.	Once locked, this follows price to secure profit.

Allow Same (Pair & Direction)	Blocks duplicate sequences on same pair.	Prevents overlapping exposure.
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🧠 This is your **profit-locking system** — how the EA exits positions while protecting equity.

4 Lot-Sizing Logic

Defines how the position size evolves — crucial for balancing recovery vs DD.

Setting	Function	Impact
Use Compounding	Enables equity-based scaling.	Lot sizes scale automatically as balance changes.
Initial Balance Threshold	Reference balance for compounding.	e.g. 100 000 → all lot math uses that as base.
Risk % for Compounding + Risk in Pips	Determines lot size dynamically.	Calculates $1R = \text{Risk\%} \times \text{Balance over PipRisk}$.
Lot Size Exponent	Multiplies each subsequent order.	1.2 = +20% each trade; > 1.5 = martingale-style.
Max Lot Size	Caps growth.	Prevents runaway exposure in long grids.
Max Running Loss (\$)	Account-level safety brake.	All positions close once open DD hits this limit.

🧠 This is the **risk engine** — decides how fast position weight builds relative to drawdown.

5 Session Control & Auto-Management

Lets you define when EA trades or rests.

Setting	Function
Close for Weekend / Day / Time	Auto-close all trades before weekend.
Trading Session (Mon-Fri)	Custom active hours (e.g. London only).

Action at End of Session	Close all / Wait for sequence / Pause.
Restart EA After Loss	Delay auto-restart after hitting DD.
Daily Profit Target / Ultimate Target Balance	Daily or total profit cap before shutdown.

🧠 This block gives **schedule discipline** — keeps you within prop-firm and market-volatility windows.

📊 6 Technical Filters (Signal Logic Layer)

These govern *entry validation* — when to start a new sequence.

Indicator	What It Does	How It's Used
RSI	Detects overbought/oversold.	Filters entries so grids start only at extremes.
EMA	Measures directional bias.	Aligns trades with or against trend.
ADX	Measures trend strength.	Avoids entering when volatility is too low.
Bollinger Bands	Detects volatility expansion or exhaustion.	Avoids chasing extended moves.

🧠 These filters are your **signal gates** — they decide “when to begin” a grid, not how it behaves after.

⚠️ 7 Risk & Equity Safeguards

Setting	Function	Analogy
Global Equity Stop Type	Absolute / Risked / %	Defines how far equity may drop.
Global Equity Stop (\$ or %)	Threshold trigger.	Auto-closes all trades.
Reset Global Equity Stop?	Clears lock after stop triggered.	Manual or automatic reset.

Min Seconds Between Trades

Enforces spacing between orders.

Prevents prop-firm violations.

🧠 This is the **failsafe circuit** — ensures one rogue sequence doesn't blow the account.

8 News Filter Logic

Setting	Purpose
Use High-Impact News Filter	Suspends trading before major news.
News Trades Action	Either close everything or just pause new entries.
Close Trade X hrs Before / Pause After	Defines blackout window around event.

🧠 Keeps the EA out of high-slippage conditions (CPI, NFP, FOMC).

9 Restart & Automation Logic

Setting	Function
Restart Next Day At / After Hours	Auto-resume after stop events.
Ultimate Target Balance	Long-term goal, shuts EA once achieved.
Restart EA After Loss	Re-enables trading after cool-down period.

In Summary

Archangel's logic stack works like this:

Signal Filters (RSI/EMA/ADX/BB)

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Entry Timing (Delay / LiveDelay / Direction)

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Grid Geometry (PipStep / Exponent)
↓
Lot Management (LotExponent / Compounding)
↓
Risk Guardrails (MaxOrders / MaxDD / GlobalStop)
↓
Exit System (LockProfit / Trailing / TP)
↓
Session + News Control (when to pause / resume)

Would you like me to give you a **diagram** (visual flow) that maps how these layers interact — from signal detection → grid build → equity protection — so you can literally see how the system behaves tick-by-tick?

You said:

no explain it more in depth, everything, like I am complete beginner, then after that go into depth, then I will ask you questions

ChatGPT said:

Awesome—let's build this from the ground up. I'll start like you've never touched an EA before, then layer in the "why" and "how to tune" for each setting. Keep this handy; it's your mental model for Archangel.

Part A — First-principles: how Archangel actually trades

1) What the EA is

Archangel is a **sequence/grid mean-reversion EA**. It doesn't place one trade and wait; it **builds a series ("sequence") of trades** as price moves against the first entry. Those trades are spaced by a **Pip Step**. When price snaps back, the **weighted average** entry (thanks to multiple positions) is easier to reach; the whole sequence **closes together** for a profit using **TP** or **lock-profit + trailing**.

Picture a fishing net (the grid) thrown below/above current price. As price wiggles into the net, the EA adds positions at planned distances. When price bounces, you pull the whole net up—banking the catch.

2) The lifecycle of a sequence (simplified)

1. **Signal filter(s) say “OK”** (RSI/EMA/ADX/BB + session/time/news rules).
2. **First trade opens** (unless you delay it).
3. **Price moves against you** → EA adds the next trade **Pip Step** away (possibly farther if **Pip Step Exponent > 1**).
4. **Lot size** of later trades can be larger (**Lot Size Exponent**) or capped (**Max Lot Size**).
5. Once the **average price** of the open bundle is within **TP** (or enough profit is locked), EA **closes the whole bundle**.
6. If **Allow New Sequence** is TRUE → it may start again when valid. If FALSE → it stops until you tell it otherwise.

3) Where drawdown comes from

DD is controlled by:

- How **dense** your grid is (**Pip Step, Exponent, Max Orders**)
- How **big** later orders are (**Lot Size Exponent/Max Lot Size**)
- How **quickly** the market trends one way (filters + sessions/news matter)
- Whether you **delay** early trades (often reduces DD)

Your goal: **profit per DD** high enough to hit your equity targets without blowing the risk budget.

Part B — Settings, explained like a beginner → then deeper

I group them by purpose. For each: **Beginner** (what it does), **Deep dive** (why it exists/how it interacts), **Use it when**, **Typical ranges** (for your M1 scalping, 0.01 lot, target ~\$60–\$80 DD per file).

1) Sequencing “power switches”

Allow New Sequence

- **Beginner:** TRUE = EA keeps starting new sequences whenever signals appear. FALSE = finish the current one and stop.
- **Deep dive:** Useful to pause expansion during news/sessions without killing the current sequence.
- **Use it when:** You want the EA to stop re-arming automatically (e.g., near Friday close).
- **Typical:** TRUE for normal; set FALSE before maintenance or manual breaks.

Magic Number

- **Beginner:** Unique ID so the EA knows which trades are “mine.”
- **Deep dive:** Critical when running multiple setfiles/pairs—prevents cross-management.
- **Use it when:** Always. Make it unique per setfile (e.g., accountID_pair_variant).
- **Rule:** Never change while trades are open.

Use Random Entry Delay

- **Beginner:** Adds a small random wait before placing a valid trade—to look unique.
 - **Deep dive:** Helps avoid copier/prop “copy patterns.” Doesn’t change expectancy.
 - **Typical:** TRUE on prop; FALSE for backtest determinism.
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2) Volatility model (ATR) & grid geometry

Use ATR for pips + ATR Period

- **Beginner:** If TRUE, your pip distances become **ATR multiples** (auto scale with volatility). If FALSE, they're fixed pips.
- **Deep dive:** ATR-based spacing keeps DD steadier across quiet vs volatile days.
- **Use it when:** You want robustness across regimes.
- **Typical:** ATR Period **4–14** on M1; start with **6**. Multipliers come from Pip Step, Lock Profit, etc.

Pip Step

- **Beginner:** Distance between entries in a sequence.
- **Deep dive:** Smaller step = faster grid buildup → higher DD, faster recoveries. Larger step = slower buildup → lower DD, slower recoveries.
- **Typical:** Fixed pips **2–4** on M1 majors or **$2.2\text{--}2.8 \times \text{ATR}$** if ATR-mode.

Pip Step Exponent (a.k.a. step multiplier)

- **Beginner:** Makes each new gap bigger than the last (if > 1).
- **Deep dive:** Reduces clustering and DD in trends; slightly slows recovery.
- **Typical:** **1.1–1.4** (gentle dispersion). 1.0 = equal gaps.

Max Pip Step

- **Beginner:** Upper cap for the distance between trades.
 - **Deep dive:** Prevents steps from exploding in trends when using exponent.
 - **Typical:** 20–40 pips on M1; or the ATR-equivalent cap.
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3) Sequence timing & entry scheduling

Delay Trade Sequence

- **Beginner:** Skip the first N trades; your “first live” trade actually occurs at level N+1.
- **Deep dive:** You let price extend first, then you join—**reduces DD**. Good with RSI divergence.
- **Typical:** 1–3 on M1 to reduce shallow chop entries.

Live Delay

- **Beginner:** Postpone several early trades and **execute them together** later.
- **Deep dive:** Bunching gives a heavier average at a better location; riskier if timed poorly.
- **Typical:** 2–3 only if you understand the impact; often 0 (off).

Combine Live Delay Trades

- **Beginner:** Merge those delayed trades into one larger ticket (cleaner).
- **Deep dive:** Same risk, fewer tickets; good for prop rules/logs.

Lot Multiplier for 1st Trade after LD

- **Beginner:** Increase the very first “real” trade size after the live-delay block.
- **Deep dive:** Gives extra weight to the best (deeper) entry; increases variance.
- **Typical:** 1.0 (off) unless you’ve tested this carefully.

Trade Direction

- **Beginner:** Long & Short, Long only, or Short only.
- **Deep dive:** Can force trend-following bias during strong regimes.

- **Typical:** Both, unless news/trend filter says otherwise.

Max Orders (Per Direction)

- **Beginner:** Hard cap on number of buys or sells.
- **Deep dive: Primary DD brake.** But fewer orders = farther TP (harder to exit).
- **Typical: 15–25** on M1 scalpers; you used **30** earlier—safe if lot sizes are small and step wide enough.

Reverse Sequence Direction

- **Beginner:** Flip buys↔sells.
 - **Deep dive:** For anti-correlation experiments or when a pair behaves better inverted.
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4) Exits & profit protection

TakeProfit (pips or ATR multiple)

- **Beginner:** Profit target for the **whole sequence** (based on average price, not each ticket).
- **Deep dive:** Too small → many closes but more churn; too big → sequences drag on.
- **Typical: 6–15 pips** on M1 majors or **0.5–1.0 × ATR**.

Lock Profit Min Trades

- **Beginner:** Don't start trailing until at least N trades are open.
- **Deep dive:** Avoids micro-profits before the grid forms.
- **Typical: 1–3.**

Lock Profit (pips / ATR multiple)

- **Beginner:** Once the sequence reaches breakeven + X, set a protective stop there.
- **Deep dive:** Acts like “break-even + buffer.” Enables trailed winners.
- **Typical:** 3–8 pips on M1 or $0.2\text{--}0.5 \times \text{ATR}$.

Lock Profit – When to check / Trailing – When to check

- **Beginner:** How often to evaluate (Every tick / Bar close).
- **Deep dive:** Every tick = responsive but noisier; bar close = stable but slower.
- **Typical:** **Every tick** for M1 scalping; **M1 close** if you want calmer logs.

Trailing Stoploss (pips / ATR multiple)

- **Beginner:** Distance the stop trails behind price once locked.
- **Deep dive:** Smaller = bank more often, risk “stop-outs”; larger = hold for bigger runners.
- **Typical:** 5–12 pips on M1 or $0.3\text{--}0.7 \times \text{ATR}$.

5) Lot sizing & compounding

Lot Size (base)

- **Beginner:** First trade size. With 0.01 minimum, this is your floor.
- **Deep dive:** Your lot ladder often controls closure speed more than TP distance.

Lot Size Exponent (LSE)

- **Beginner:** Each subsequent trade grows by this factor.
Example: $0.01 \rightarrow 0.012 \rightarrow 0.014 \rightarrow \dots$ if $\text{LSE}=1.2$.
- **Deep dive:** **Biggest DD lever.** Higher LSE = faster recovery but **steep DD** in trends.

- **Typical:** 1.05–1.25 (gentle). Avoid >1.5 on prop.

Max Lot Size

- **Beginner:** Hard cap on any individual ticket size.
- **Deep dive:** Safety net when LSE is >1; prevents runaway size.

Use Compounding + Initial Account Balance Threshold + Risk% + Risk in Pips + Max Lot Size for Compounding

- **Beginner:** Auto-sizes lots vs account/equity and a defined stop (or “risk in pips”).
 - **Deep dive:** Good for swing systems with explicit SL. For grid EAs, use with care.
 - **Typical:** Often OFF for your M1 sets (use fixed 0.01 + gentle LSE).
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6) Session control & automation

Close for Weekend / Day/Time to Close & Restart

- **Beginner:** Auto-flat and pause before weekend (or whenever you choose).
- **Deep dive:** Avoids gaps/slippage. Good for prop firm rules.
- **Typical:** Friday 20:00–21:00 server close; Monday restart 01:00–03:00.

Trade Custom Times (Mon–Fri) + Action at End of Session

- **Beginner:** Only trade inside your chosen windows (e.g., London).
End-of-session: close all / wait to close / pause.
 - **Deep dive:** Risk drops if you avoid rollover/spread spikes and news hours.
 - **Typical:** London + first half of NY; off during rollover and late NY.
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7) Account-level safety

Max Running Loss (\$)

- **Beginner:** If open-trade DD hits this amount → close everything.
- **Deep dive:** This is your **per-setfile** equity brake. You're using **\$69–\$80** caps—good.
- **Typical:** Set to your budgeted DD (e.g., \$69).

Global Equity Stop Type + Global Equity Stop + Reset

- **Beginner:** Portfolio-wide kill-switch by absolute equity, risked amount, or %.
- **Deep dive:** This is your **master fuse** across all setfiles.
- **Typical:** For your 10k → **Absolute 9,580** for ~4.2% buffer; or **Risked 420**.

Restart EA After Loss (next day / after N hours)

- **Beginner:** Cool-down before re-arming the strategy.
- **Deep dive:** Prevents re-entry into the same trend storm.

Min Seconds Between Trades

- **Beginner:** Enforces a minimum time gap; some prop firms require this.
 - **Typical:** **30–60s** if needed; otherwise 0.
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8) Signal filters (when to even start a sequence)

RSI (TF, Period, Overbought level)

- **Beginner:** Start sells only when RSI high; buys when RSI low (mirrored levels).

- **Deep dive:** Good for counter-trend mean reversion; combine with **Delay Trade Sequence** for divergence.
- **Typical:** TF **M5–M15**; Period **7–14**; Levels **70/30** or **80/20** for stricter entries.

EMA (TF, Fast–Mid–Slow, Trend Rule, Double Check)

- **Beginner:** Use moving averages to define trend. Trade with trend only, or allow both in flat markets.
- **Deep dive:** Reduces “fighting trend” risk; **Double Check** ensures the first real trade respects the EMA rule after delays.
- **Typical:** TF **M30/H1**; EMAs like **5–20–60** or **8–21–60**.

ADX (TF, Period, Threshold, Trend Rule, Double Check)

- **Beginner:** Measures trend strength.
Above threshold = strong trend; below = range.
- **Deep dive:** You can forbid countertrend grids in strong trends (saves DD).
- **Typical:** Threshold **20–25**; Period **7–14**.

Bollinger Bands (TF, Period, Deviation, Mode)

- **Beginner:** Avoid chasing extremes or only take extremes counter-trend.
- **Deep dive:** Pairs well with grid: **Avoid Extreme** in trend mode; **Only Extreme Counter Trend** in ranging regimes.
- **Typical:** TF **M30/H1**; Period **20**; Dev **1.5–2.0**.

9) News filter

Use High Impact News Filter + Actions + Before/After windows

- **Beginner:** Pause or close around major news.
 - **Deep dive:** Slippage + spread spikes are grid killers. Safer to stand down.
 - **Typical:** Close 1h before, resume 2–4h after for USD/red events; or “manage open sequence only” if you must.
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Part C — How to tune for your goal (M1, 0.01 lot, ~\$69–\$80 DD per setfile)

1. **Start simple:** ATR OFF (fixed pips), **Pip Step 2.2–2.8**, **Exponent 1.1–1.25**, **Max Orders 18–24**, **LSE 1.10–1.20**, **Max Lot cap reasonable** (e.g., 0.10–0.20 on a 10k with 0.01 base).
 2. **Reduce early entries:** **Delay Sequence = 1–2**; **Live Delay = 0** initially.
 3. **Exit protection:** **Lock Profit 4–8 pips**, **Trailing 6–10 pips**, **check Every tick**.
 4. **Filters:** At minimum **EMA trend rule** (Avoid opposite trend in strong trend), **ADX threshold 20–25** to avoid strong one-way trends.
 5. **Kill-switches:** **Max Running Loss = \$69** per setfile; **Global Equity Stop** around **–\$420** (your 4.2% account DD).
 6. **Sessions:** Trade **London→early NY**, avoid **rollover** and **late NY**.
 7. **Validate:** Backtest 3 months + forward 1 month. Keep **Profit/DD ≥ 2.4** , **PF ≥ 1.8** , **max DD $\leq \$85$** .
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Part D — Quick “what to change when ...”

- **DD too high**
 ↑ Pip Step, ↑ Pip Step Exponent, ↓ Max Orders, ↓ LSE, enable EMA/ADX filters, Delay

Sequence +1

- **Profit too low but DD small**
↓ Pip Step slightly, ↓ Exponent, ↑ LSE a touch (e.g., 1.10→1.15), tighten Lock Profit/Trailing so it cycles more
- **Many small stop-outs**
Increase Lock Profit & Trailing distances; check “When to check” to bar close; widen session filter
- **Chopped to death in range**
Add **Delay Sequence**, use **BB Avoid Extreme**, RSI stricter (80/20), ATR mode ON with small multipliers
- **Killed by trends**
Enable EMA trend rule + ADX filter; reduce Max Orders; raise Pip Step Exponent; add News filter around trend-events